Suryasri Shanmukha Aketi

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TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, SQL, C, Swift
ML and Computer Vision: TensorFlow/Keras, OpenCV, NumPy, SciPy; U-Net, semantic segmentation
Tools and Development: React, Git/GitHub, Linux, SLURM, ROAR, GPU computing, Python virtual environments

EDUCATION

Penn State University State College, PA
Bachelor of Science in Computer Science Expected May 2027

WORK EXPERIENCE

Penn State University May 2026 - Aug 2026
Machine Learning Research Intern | Industry-supported research

- Trained and evaluated TensorFlow/Keras U-Net models for grain-boundary segmentation in microscopy images, comparing train/validation splits using precision, recall, F1, IoU, and prediction overlays.
- Developed Python/OpenCV pipelines for Sobel, Canny, phase congruency, multiscale Shannon entropy, 2D entropy, and Tsallis entropy to assess boundary evidence and annotation quality.
- Benchmarked detectors across six annotated images using threshold sweeps and one-pixel-tolerant evaluation; phase congruency detected an average 8.32% of annotated boundary pixels missed by Sobel.
- Ran GPU training on Penn State's ROAR HPC cluster using Linux, Bash, and SLURM; managed Python environments, monitored experiments, and troubleshot dependencies.
- Created and analyzed PNG/SVG annotation masks and documented quantitative results and visual comparisons for Professor Xing Wang and PhD researcher Aiden Ochoa.

Wyntek Solutions Inc. May 2024 - Aug 2024; Jun 2025 - Aug 2025
Machine Learning Intern | Monmouth Junction, NJ

- Cleaned and preprocessed datasets and wrote supporting code to assist data engineers with predictive modeling.
- Built and trained baseline machine learning models, tuned hyperparameters, and ran evaluation scripts under a data engineer’s mentorship.
- Tested model behavior and helped integrate models into staging and production pipelines, collaborating with data engineers through daily stand-ups.

CERTIFICATIONS

Machine Learning Specialization | DeepLearning.AI and Stanford Online (Coursera) | Completed
Courses: Supervised Machine Learning: Regression and Classification; Advanced Learning Algorithms; Unsupervised Learning, Recommenders, Reinforcement Learning.

CS50: CS50P (Programming with Python), CS50AI (Artificial Intelligence with Python), and CS50x (Computer Science)

PROJECTS

LocalScribe | On-Device Dictation & Transcription

- Built a macOS menu-bar application using Swift, SwiftUI, and WhisperKit for local speech recognition, live transcription, and automatic text insertion into other applications.
- Implemented app-specific output modes, spoken-code formatting, microphone calibration, and searchable transcript history with audio playback and export.
- Added local noise reduction, audio normalization, silence detection, and a hands-free stop command.

TaskBridge | AI Desktop Assistant

- Designed a modular macOS assistant architecture separating AI reasoning and task planning from desktop interaction through a dedicated execution bridge.
- Defined command and event interfaces for screen capture, application control, and file operations, incorporating permission checks and action verification.
- Specified a policy layer to validate action requests before passing them to the desktop execution bridge.